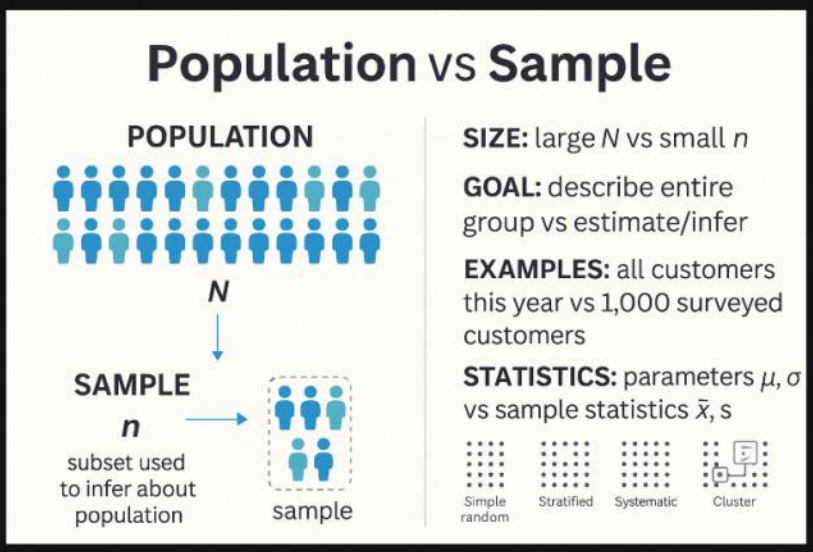
CN-Stats-Lec01-Introduction to Statistics & Types of Data and Data Collection

Introduction to Statistics

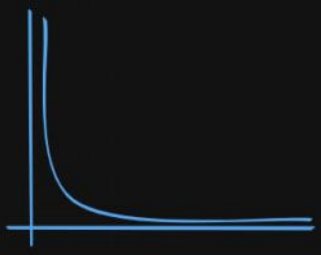
Session Objectives:

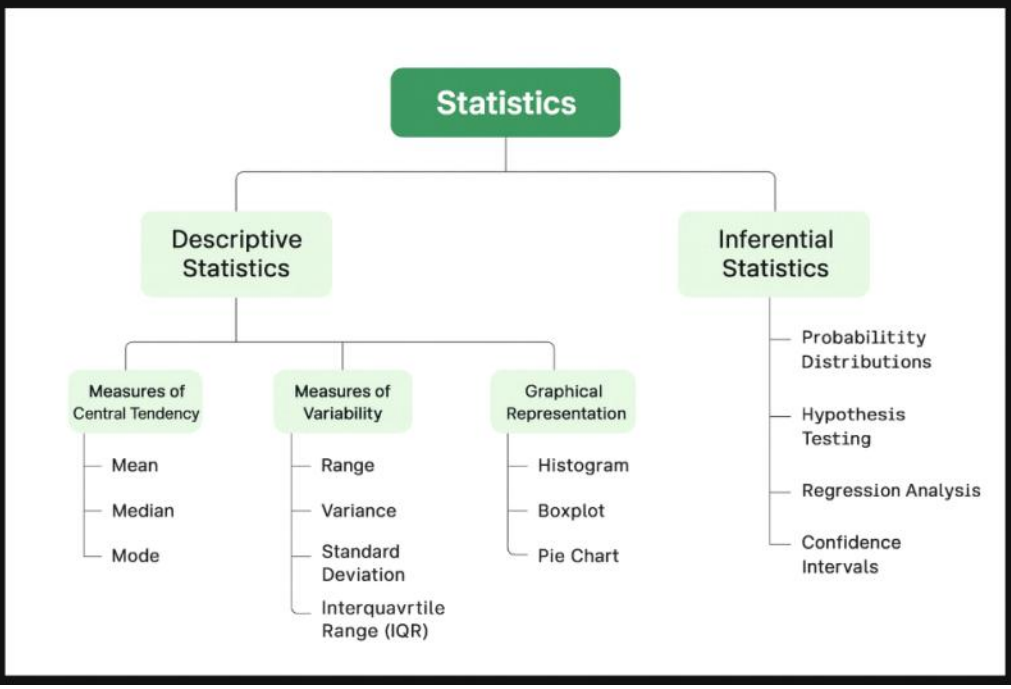
- Understand the course structure and objectives.
- Explain fundamental concepts (central tendency & variability).
- See how statistics are applied in real life.
- Understand how statistical methods help decision-making.
- Get familiar with Python libraries (SciPy, Scikit-learn, StatsModels).



Hypothesis Testing
H0: Null Hypothesis &
H1: Alternative Hypothesis

if-else





Univariate Analysis

Categorical Column
Continuous Column

Bivariate Analysis

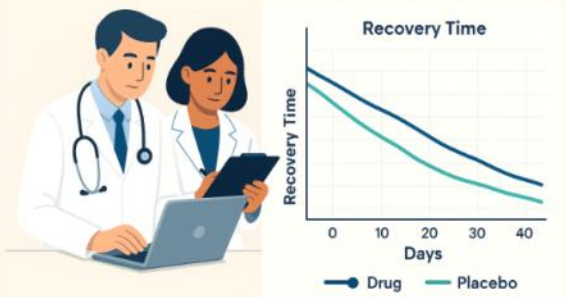
Categorical Column
Continuous Column

DESCRIPTIVE vs INFERENTIAL STATISTICS: KEY DIFFERENCES

DESCRIPTIVE	INFERENTIAL
• Summarizes and organizes data	• Draws conclusions about data
• Entire dataset	• Sample of data
• Tables, charts and graphs	• Probability statements
• Measures of central tendency and spread	• Relationships and predictions
• "What does the data look like?"	• "What conclusions can be drawn?"
• Mean, median, mode, range, etc.	• p-values, confidence intervals, etc.
	• Generalize data

Real-World Examples:

CLINICAL TRIAL COMPARISON



Healthcare

Compare new drug vs placebo using trial data.

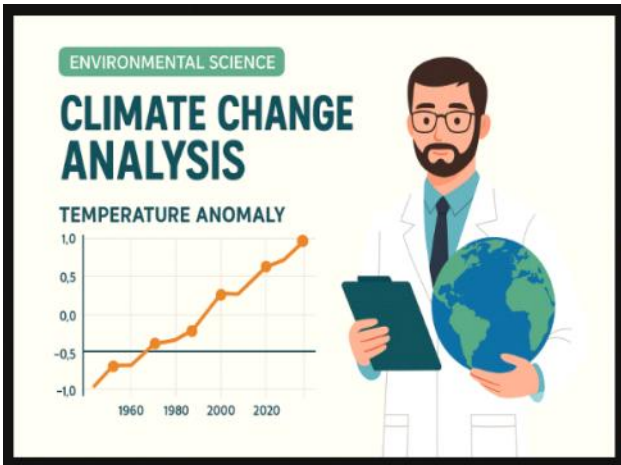
Decide if the improvement is real (not just chance).

PREDICTING FUTURE SALES

Business & Marketing

Use past sales to predict future sales.

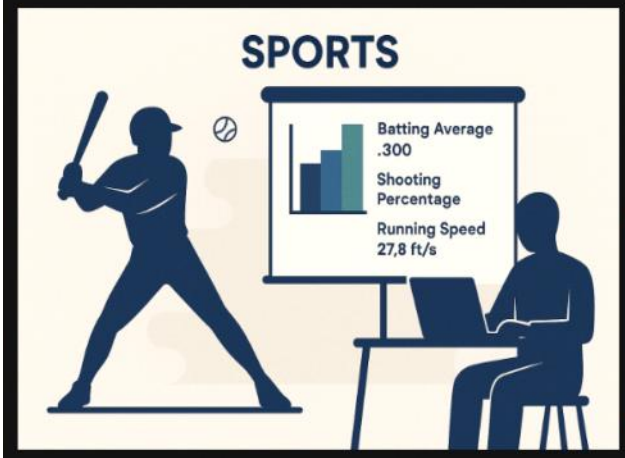
Plan inventory/ads based on the forecast.



🌍 Environmental Science

Track temperature over years to spot trends.

Forecast future climate patterns.



🏆 Sports

Analyze player/team stats (e.g., averages, speeds).

Improve training & game strategy.

👥 Social Sciences

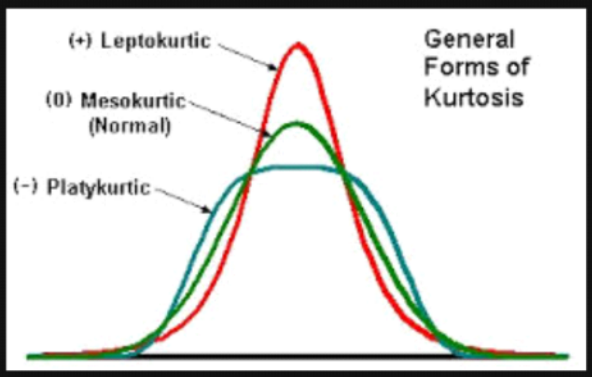
Use surveys of a sample to understand the population.

Find relationships (e.g., education ↔ employment).

Measures of central tendency

$$\text{Skew} = \frac{n}{(n-1)(n-2)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^3$$

$$\text{Kurtosis} = \left\{ \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum \left(\frac{x_i - \bar{x}}{s} \right)^4 \right\} - \frac{3(n-1)^2}{(n-2)(n-3)}$$



$$\Pr(\mathrm{H}|\mathrm{E}) = \frac{\Pr(\mathrm{E}|\mathrm{H})\Pr(\mathrm{H})}{\Pr(\mathrm{E}|\mathrm{H})\Pr(\mathrm{H}) + \Pr(\mathrm{E}|\text{not H})\Pr(\text{not H})}$$

The chance evidence is real (supports a hypothesis) is the chance of a true positive among all positives (true or false)

stem - and - leaf method

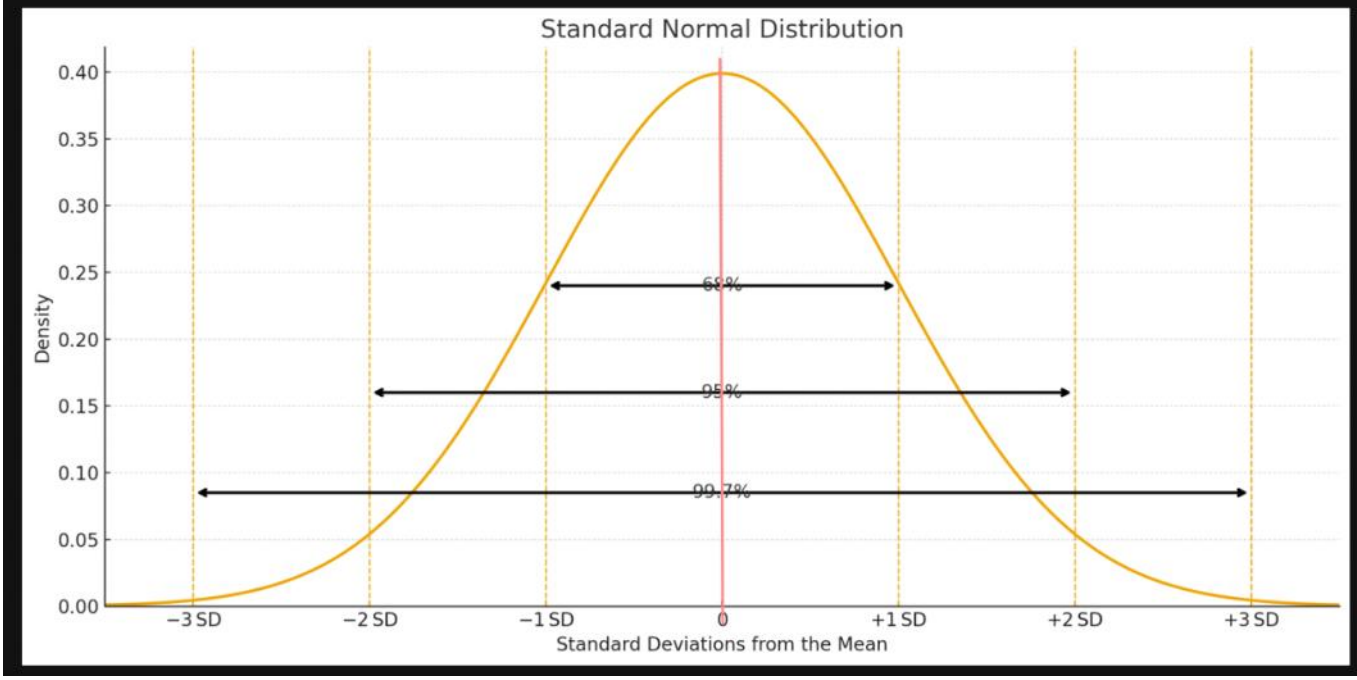
	3,5,7,9	3,5,7,9
6	7,8	67,68
7	1,2,3,4	71,72,73,74

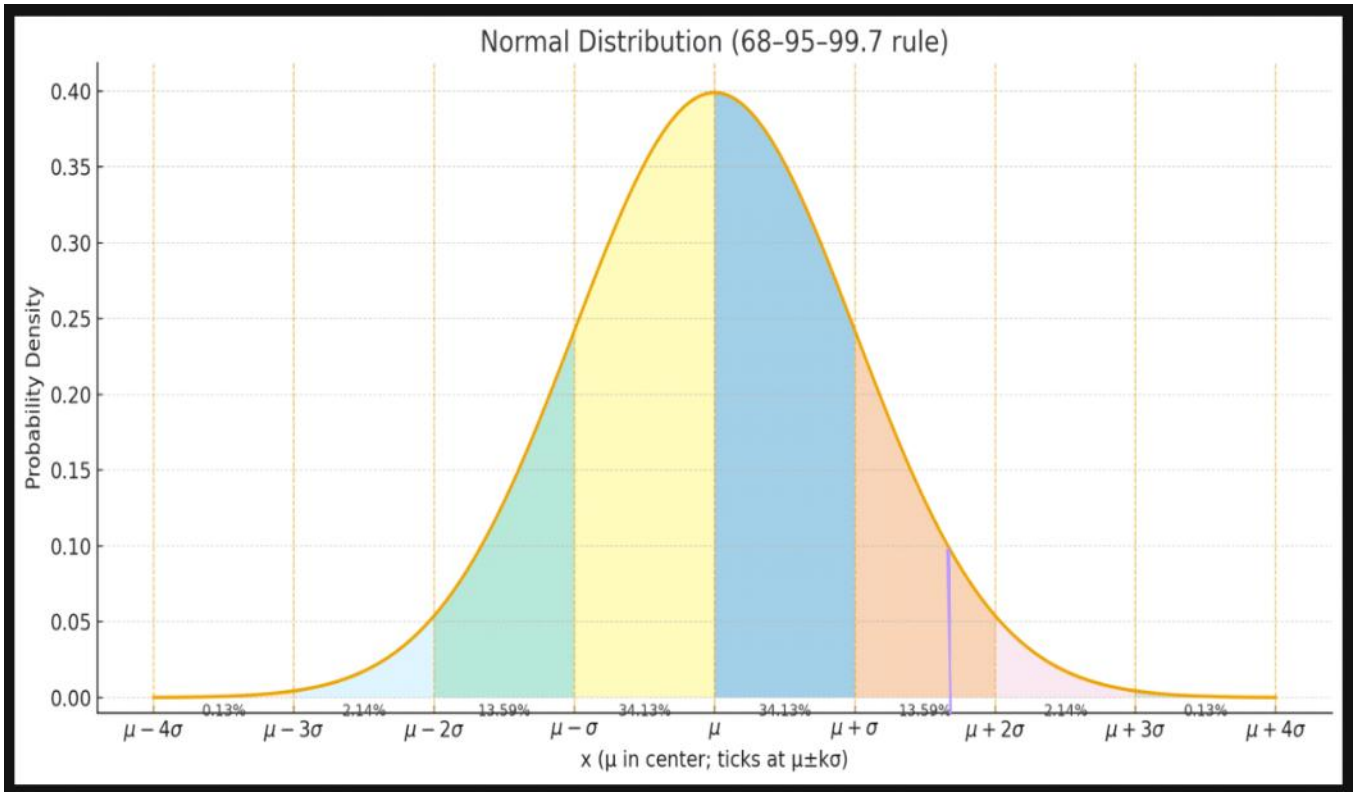
```
43, 47, 48, 51, 52, 52, 54, 56, 61, 62, 67, 68, 69, 70, 73

Stem: the tens place (4, 5, 6, 7)
Leaf: the ones place (3, 7, 8, ...)
Stem | Leaf
---------------
4    | 3 7 8
5    | 1 2 2 4 6
6    | 1 2 7 8 9
7    | 0 3
```

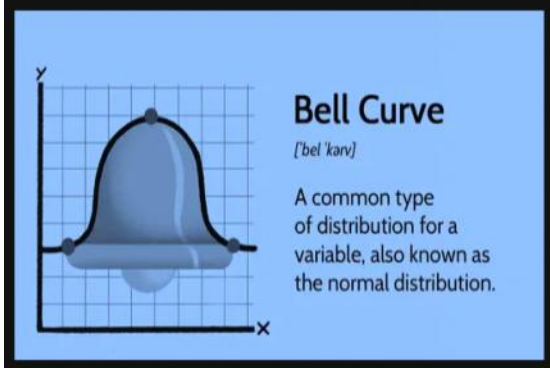
Types of Data and Data Collection

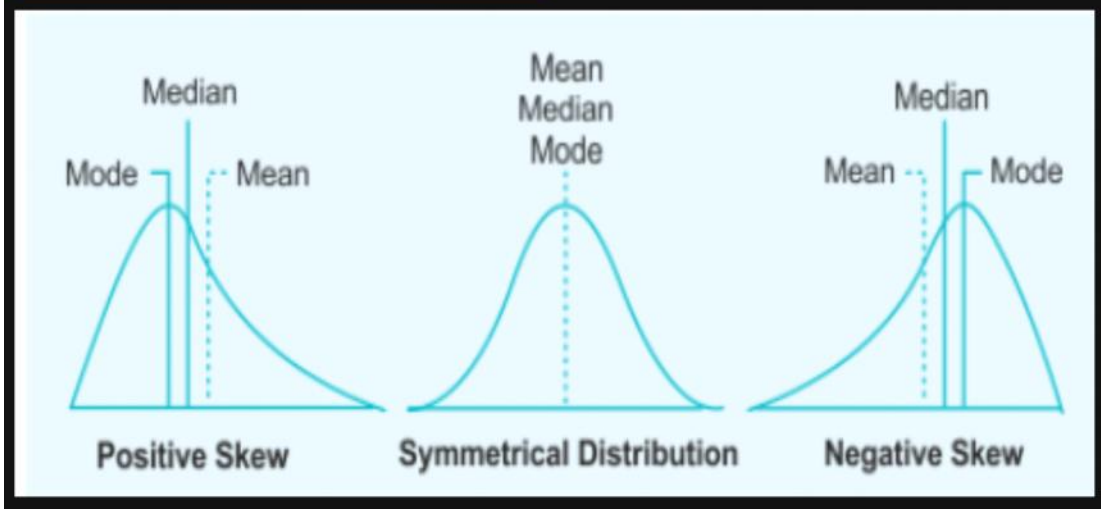
🎯 Session Objectives:

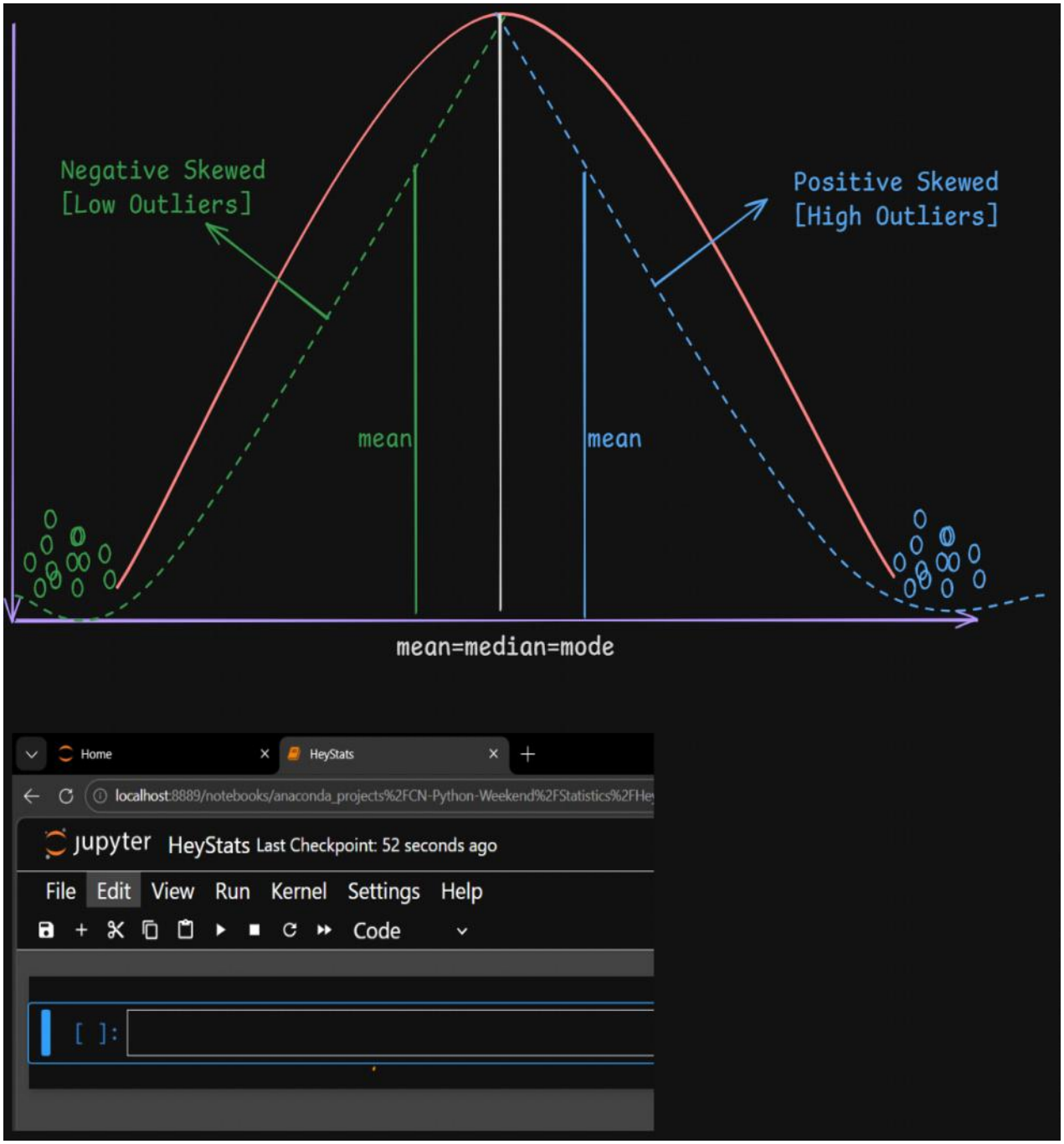
- ✅ Get familiar with Python libraries (SciPy, Scikit-learn, StatsModels).
- 🔍 Identify and categorize different types of data.
- 🏷️ Distinguish between Nominal and Ordinal data.
- 🔢 Explain Interval and Ratio data.
- 📐 Differentiate between Discrete and Continuous data.
- 📊 Understand data collection methods & sampling techniques.




Normal Distribution (68–95–99.7 rule)
Probability Density
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.13%
2.14%
13.59%
34.13%
34.13%
13.59%
2.14%
0.13%
μ − 4σ
μ − 3σ
μ − 2σ
μ − σ
μ
μ + σ
μ + 2σ
μ + 3σ
μ + 4σ
x (μ in center; ticks at μ±kσ)


Y
X
Bell Curve
['bel 'kərv]
A common type of distribution for a variable, also known as the normal distribution.


Median
Mode
Mean
Positive Skew
Mean
Median
Mode
Symmetrical Distribution
Median
Mean
Mode
Negative Skew


Negative Skewed
[Low Outliers]
Positive Skewed
[High Outliers]
mean
mean
mean=median=mode
Home
HeyStats
localhost:8889/notebooks/anaconda_projects%2FCN-Python-Weekend%2FStatistics%2FHe
jupyter HeyStats Last Checkpoint: 52 seconds ago
File Edit View Run Kernel Settings Help
Code
[ ]: